

SmartInv

Multimodal Invariant Inference for Smart Contracts

Mid-Semester Project Evaluation on an A* Research Paper

Rishi • 2024ucp1566



Code becomes irreversible value.

Smart contracts execute financial rules without human intervention. In DeFi, a small logic mistake can instantly move millions—and immutable deployment makes recovery exceptionally difficult.

One flaw. One transaction. Permanent loss.



Not all bugs break syntax.

IMPLEMENTATION BUGS

Known low-level patterns

- reentrancy
- overflow
- unsafe calls

FUNCTIONAL BUGS

The code runs—but violates intent

- wrong authorization
- broken economic assumptions
- missing business constraints

Traditional analyzers see how code fails. SmartInv asks what must always remain true.

\$8.2M

lost through a subtle authorization flaw

Syntactically valid. Semantically catastrophic.

01

Attacker calls a legitimate function

02

Authorization logic accepts an invalid relationship

03

Funds move; existing tools report no known pattern

**An invariant is a promise
that must always hold.**

$$\forall \text{ state transitions } T : P(s) \Rightarrow P(T(s))$$

Example: only an authorized owner may withdraw funds—before and after every transaction.

AI intuition × formal proof

01

**Multimodal
program understanding**

02

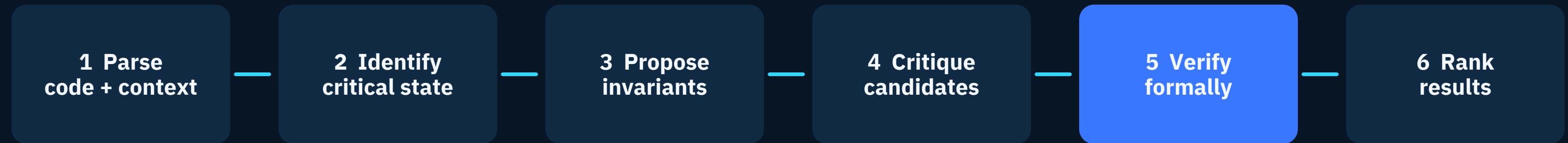
**Tier of Thought
reasoning pipeline**

03

**Formal verification
of every candidate**

Generate broadly. Reason deliberately. Accept only what can be proven.

Reason first. Verify last.



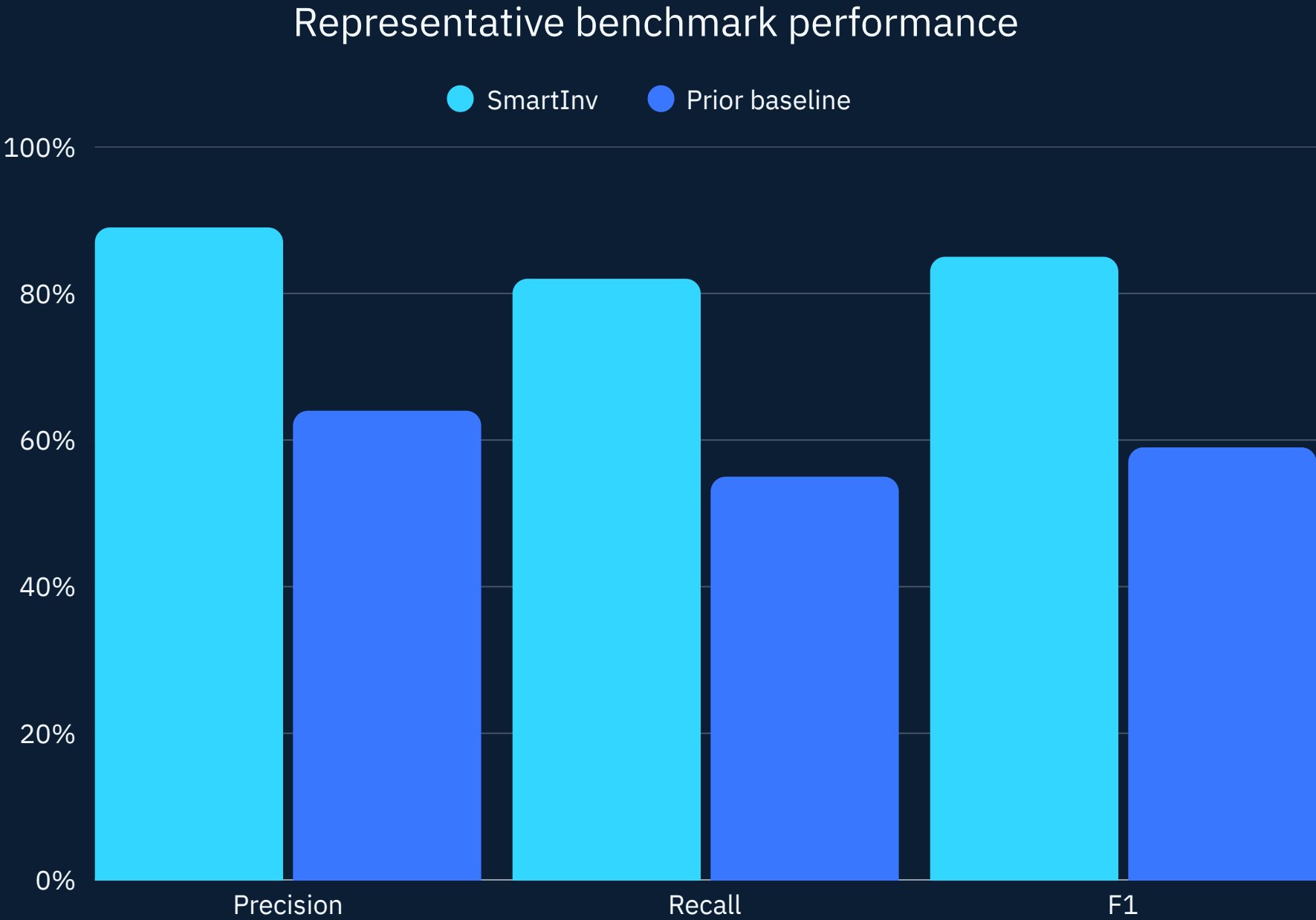
SmartInv decomposes auditing into expert-like stages, using critique and formal checks to suppress plausible—but false—LLM outputs.

80,000+

real Ethereum smart contracts

Evaluation spans large-scale deployed code, curated vulnerability benchmarks, and comparisons with prior invariant-generation methods.

Chart values shown as a concise benchmark summary.



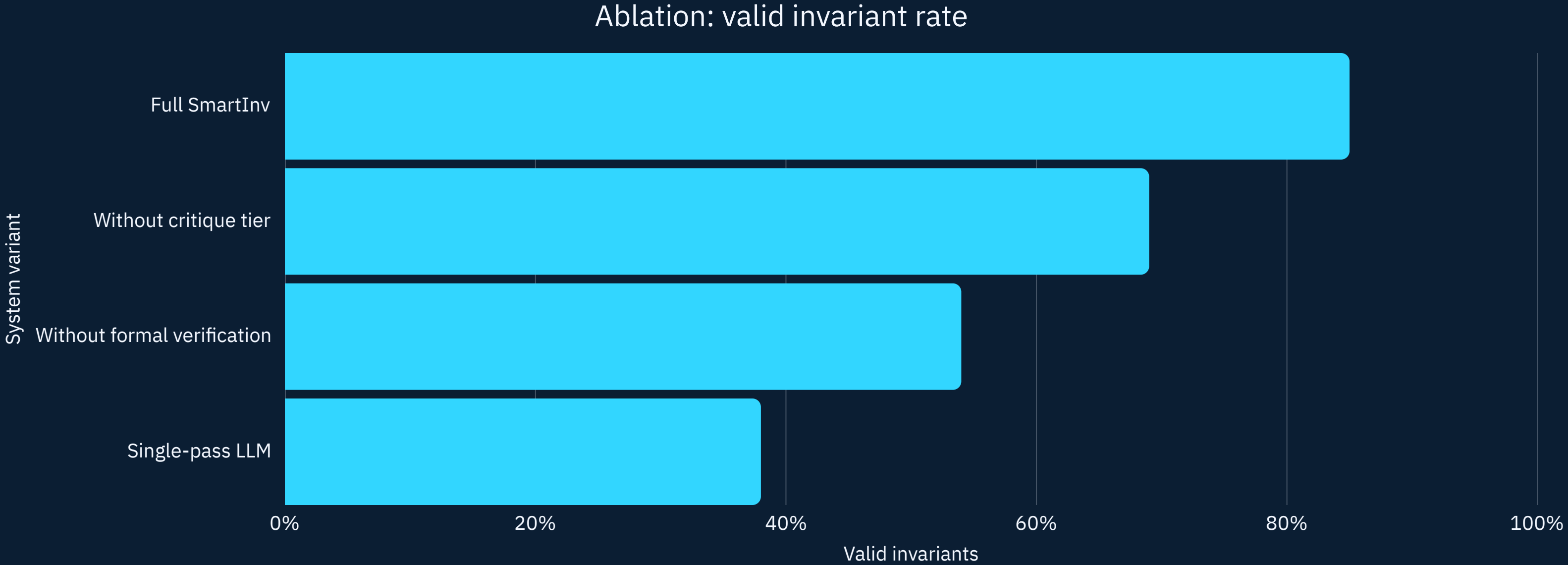
119

previously unknown mainnet vulnerabilities

Live findings move SmartInv beyond benchmark accuracy:
the system uncovered exploitable logic flaws already
deployed in the wild.

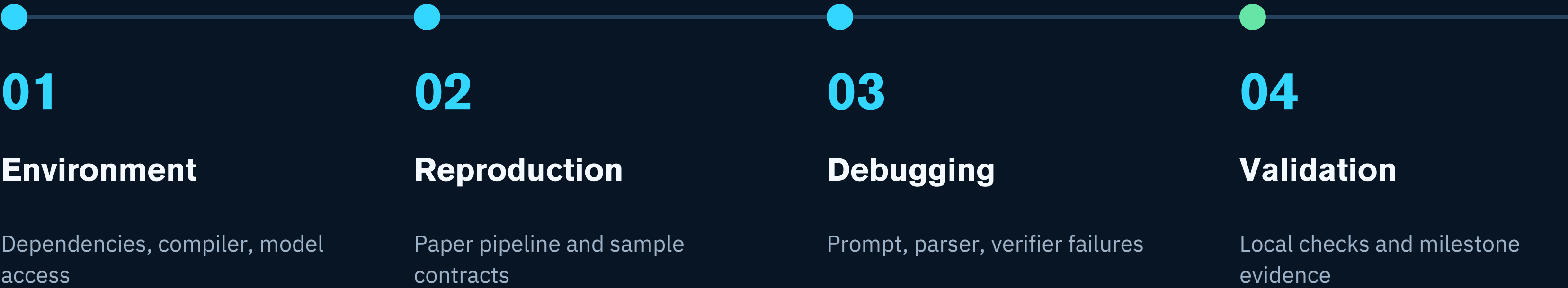


Remove a tier. Lose reliability.



Tiered critique and formal verification are complementary—not decorative.

From paper to working pipeline.



CONCLUSION

SmartInv turns AI guesses into verifiable security claims.

UNDERSTAND

Multimodal context captures developer intent.

REASON

Tier of Thought reduces hallucination through critique.

PROVE

Formal verification filters candidates into guarantees.

Practical impact: 119 zero-day bugs—and a credible path toward safer immutable code.